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A new species of Pipunculidae (diptera) from Indo-Nepal border of Champaran district, Bihar, India

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Abstract : The paper describes and illustrates a new Pipunculid (Big headed fly) species *Pipunculus cephalops* Pokharensis, sp. nov. The specimens were collected from localities in various places of Indo-Nepal border of Champaran district which has humid, mountainous, low topical vegetation. The new species resembles *Pipunculus cephalops deminitens* Hardy but can be differentiated in its smaller body, proleural fan with 4-5 bristles, 1st abdominal segment with 3 stout black bristles on both sides and piercer of ovipositor reaches upto 4th abdominal segment.

Keywords: Big headed fly, Diptera, *Pipunculus cephalops deminitens*, *P. cephalops* Pokharensis sp. nov.

INTRODUCTION

The Pipunculids belongs to the family **Pipunculidae** and of the order **Diptera** with a worldwide distribution. It is a sister group of the flower flies (Syrphidae and Platypezidae) but can easily be differentiated by their wing venation, the cell r4+5 being open and the vena spuria being absent **(Papp *et al.* 2000)**. They have affinities with Conopidae **(Cumming *et al.* 1995).** Individual flies vary in body length from 1.5 mm to 5 mm and can be distinguished by their large spherical or hemispherical head which is extremely mobile and composed almost entirely of compound eyes. These flies are therefore called **"Big-headed Flies."**They live generally in shades of herbs, shrubs, in grasses, and garden in hilly places.The life history is also interesting as they are endoparasites of various families of order Homoptera including Cicadellidae, Delphacidae, Fulgoridae **(Ferrar, 1987)**and adult Tipulidae**(Koenig & Young, 2007)**.A total of 1400 species of Pipunculidae from the world are known (**Rafael & Skevington, 2010).** As far as the taxonomy of these flies is concerned, it is largely neglected in India & Nepal. Hence the aim of our study is to reveal species richness of this area. The present study is an extension of this aspect of taxonomy, with description of a new species. As these flies are endoparasites of rice leaf hoppers and plant hoppers. Therefore, present work can be of great help for such future agroadvancement in India & Nepal.

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MATERIALS AND METHODS

The collection of flies was made from various places of Indo-Nepal border i.e. Valmikinagar, Ramnagar, Hetauda and Pokhara. Hand sweeping net method were used to collect flies.These flies were generally collected by sweeping randomly over flowers, garden, grass, herbs, shrubs and rice field in sunshine and non windy day. As these insects are rarely foundtherefore in 3 to 4 hours sweeponly 1 to 2 insects were collected and sometimes no insect was found, so collection was not possible. Dissections were done under the binocular microscope. For temporary mounting, glycerol was used while permanent preparation were made after passing the material through the alcohol series and finally mounted in Canada balsam. Diagrams were made with the help of camera lucida. The descriptions of the new species are based on the morphological characters proposed by**Cammerson, 1974** and according to the classification adopted by**Hardy,**

1972.

RESULTS AND DISCUSSION

A. Key to the species

1. 3rd section of coasta very short compared to 4th section and lacking a distinct stigma……..Tomosvaryella…..……..…2
- 3rd section of coasta with brown stigma usually equal or longer than 4th section & propleural fan absent……Eudoryllas…… 4
- 3rd section of coasta with brown stigma usually equal or longer than 4th section & propleural fan present……Pipunculus (cephalops)…….....……6
2. Wings entirely hyaline……..……3
- Wings not entirely hyaline……...4
3. Hypopygium about one half of the 5th abdominal tergum……..T. nitens Brunette
4. Third antennal segment acute. Male hypopygium with an apical membranous area and with a membranous protrusion from the apex……..Eudoryllas discors Hardy
- Third antennal segment not acute……....……..5
5. Third antennal segment acuminate. Male hypopygium with a prominent cleft extending longitudinally down the right side, surstyle broad, left surstyle has a narrow pointed end……..Eudoryllas distocruciator Hardy
- Third antennal segment more acuminate. Abdomen curved and body small oval. Arista and Ovipositor smaller……...Eudoryllas ovatum sp. nov.
6. Small bodied, 3rdantennal segment acute apex, 3rdcoastal section slightly more than half of 4thcoastal section. Piercer straight. Cross vein r-m situated at 2/5th of discal cell……Pipunculus cephalops deminitens Hardy
- Body smaller, 1/3rd propleural fan with 4-5 bristles, 1st abdominal segment with 3 stout black bristles on both sides, Piercer of ovipositor straight , cross vein r-m situated at1/3rd of discal cell. 3rd antennal segment short obtuse……*Pipunculus (cephalops)* Pokharensis, sp. nov.

B. Description of *Pipunculus (cephalops)* pokharensis, sp. nov. (Figs. 1-3)

Female: Small bodied, yellowish black.

Head: Hemispherical as broad as long, junction of the compound eye rather long about one half longer than frontal triangle. Frons- deep salivery grey pubescent, First two antennal segment brown, 3rd segment is pale yellow short obtuse apically. Arista black having double length to the 3rd segment.

Thorax: Mesonotum- Sub shining black brown with brown dusty surface having minute hairs sparsly distributed. Proleural fan with 4-5 bristles. Halter yellow with brown knob.

Wing: Hyaline highly tinged with pink brown, stigma fills the apical 2/3rd of 3rd coastal section, 3rd coastal section is slightly more than half as long as 4th. Cross vein marking of the base of the stigma very prominent covers about 3/4th as long as 4th section. Cross vein r-m is situated at basal 1/3rd of discal cell.

Legs: Predominantly yellowish brown. Coxa- brown black, Trochanter- yellow. Front and mid femora yellow shining posteroventrally. Hind femora yellow but black brown tinge on posterolateral apex. Each with well developed anteroventral and posterovental black spicules. Tibia yellow with both ends hyaline having dispersed rows of short pale hairs. Tarsi yellow, last tarsal segment slightly darkened pulvilli well developed longer than the segment. Claw equal to the size of pulvilli.

Abdomen: Shining black with sparse brownish dust. 1st abdominal segment with 3 stout black bristles on both sides. A bilateral rows of minute spars setae extend downward dorsolaterally and are in between is bare. Ovipositor with black base, globose in shape, Piercer brownish yellow slender reaching upto the margin of 4th segment. It is slightly straight and its tip is slightly curved upwards.

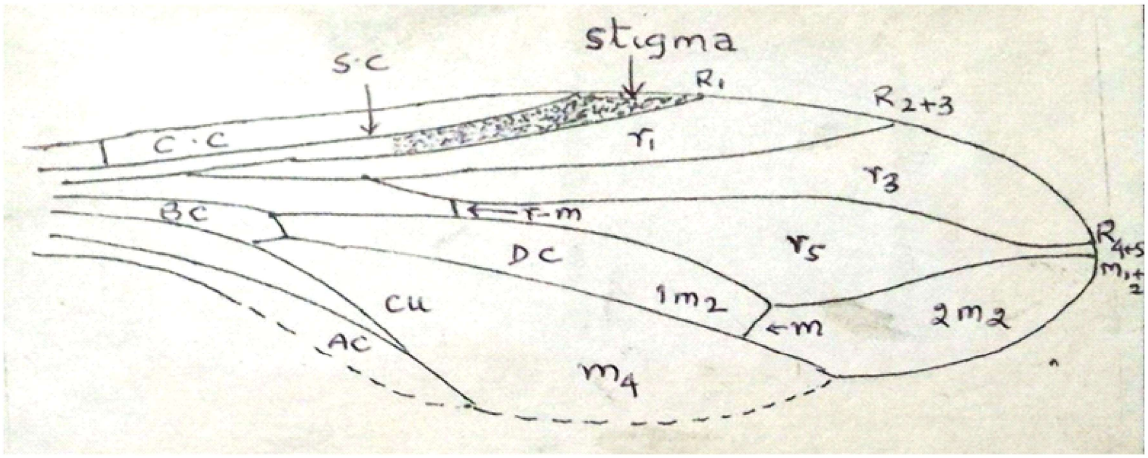
Body length: 2mm; wings: 3mm.

Specimen examined: *Holotype*: $o_{+,}$ Pokhara (Fish Tail Hotel Garden), Nepal, 28.03.2015, Coll. Shailendra Kumar Amogh.

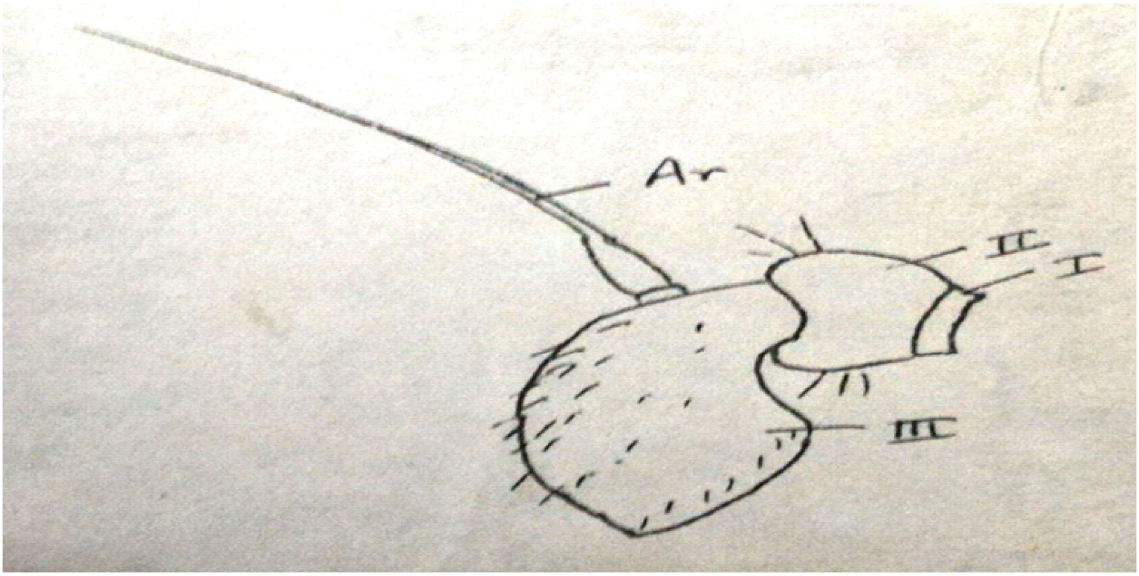
Etymology: The species named after its type locality.

REMARKS

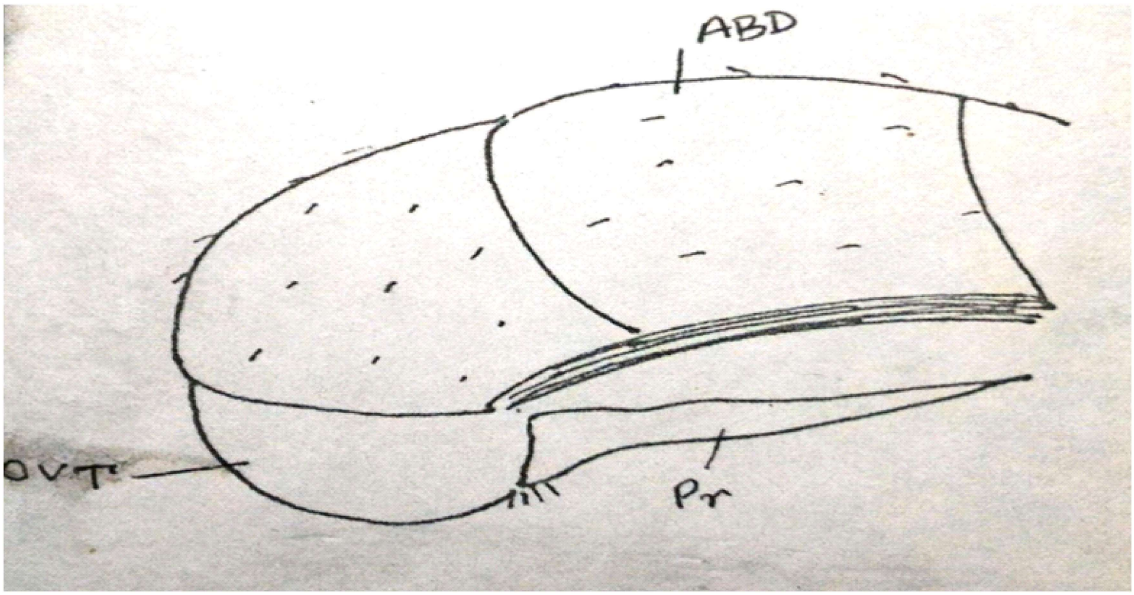
The new species is closely related to *Pipunculus (cephalops)* deminitens Hardy, 1966. However, it differs in having (1.) Body smaller, (2.) Propleural fan with 4-5 bristles, (3.) 1st abdominal segment with 3 stout black bristles on both sides, (4.) Piercer of ovipositor reaches upto 4th abdominal segment.



1



2



3

Fig. 1-3. Pipunculus (cephalops) Pokharensis, sp.nov. 1. Wing. 2. Antenna. 3. Female genitalia

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